

Smart Shortlisting Engine • System Architecture & One-Page Report

1. EXECUTIVE SUMMARY & PROBLEM SOLVED

Traditional recruitment relies either on rigid keyword regex (which disqualifies candidates for minor vocabulary variations) or blind LLM prompting (which hallucinates unexplainable scores). **InternLoom AI** resolves this with an explainable **dual-engine hybrid ranking architecture** that deterministically fuses **local vector semantic embeddings (BAAI/bge-small-en-v1.5 via scikit-learn cosine similarity)** with **multi-tier lexical term saturation (BM25 + TF-IDF)**, wrapped in an Apple-inspired 5-stage recruitment dashboard with automated HR bias mitigation.

2. TEAM MODULES & TECHNICAL DIVISION OF WORK

Member 1 (You)	Friend 1	Friend 2
Semantic AI Engine	Ingestion & Keywords	Recruiter UI & Experience
Python / FastAPI (Port 8000)	PDF Parser & BM25 Matcher	React 19 + Express (Port 3000)
- BAAI/bge-small-en-v1.5 local embeddings (no external LLMs). - FastAPI lifespan single-startup loading. - scikit-learn cosine similarity matrix. - Deterministic 0.00–100.00 score scaling. - Pure contract: POST /semantic-rank	- PyMuPDF multi-page PDF text extraction. 8-category skills.json ontology DB. - rank_bm25 term saturation retrieval. - Sublinear TF-IDF n-gram similarity. - Levenshtein typo & messy date recovery.	5-stage Apple-style recruiter flow. - Interactive weight slider (10% to 90%). - Bias Scanner for exclusionary JDs. - Side-by-side head-to-head compare. - AI Recruiter Q&A chat assistant drawer.

3. MATHEMATICAL DUAL-ENGINE FORMULATION

Score(Resume, JD) = [Wkeyword × SCOREBM25 / Ontology] + [Wsemantic × SCOREBGE-Cosine]

Placement officers dynamically adjust weights via the UI slider. The keyword engine evaluates hard technical constraints and BM25 term frequency, while the semantic engine computes contextual vector proximity using cosine distance: $\cos(\theta) = (\mathbf{u} \cdot \mathbf{v}) / (||\mathbf{u}|| \cdot ||\mathbf{v}||)$.

4. COMPLETE PLATFORM FEATURE CATALOG

Resilient Ingestion: Tolerates typos ("reaktjs" → React), parses non-standard dates ("Summer '24"), and infers missing headers. JD Bias Scanner: Detects university tier elitism ("Tier 1 only") and experience inflation, providing 1-click inclusive revisions.	Head-to-Head Comparison: Modal with score differentials, domain breakdown (Frontend, Backend, DB, DevOps), and unique skill tags. Grounded AI Recruiter Q&A: Natural language query drawer answering recruiter questions strictly grounded in candidate evidence.
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5. PRODUCTION ENDPOINTS & SERVICE CONTRACTS

Endpoint	Method	Port	Input & Contract Output Description
/semantic-rank	POST	8000	Member 1 spec: Accepts JD & resumes JSON → returns sorted [{"name": "Alice", "semantic_score": 91.42}].
/api/upload-and-rank	POST	8000	Full pipeline: Accepts PDF files + JD text → PyMuPDF parsing → hybrid scoring → complete leaderboard.
/health	GET	8000	Verifies FastAPI service health and confirms BAAI/bge-small-en-v1.5 is loaded in RAM.
/api/analyze-jd-bias	POST	3000	Audits JD for exclusionary language, computes bias score, and provides inclusive rephrasing.
/api/recruiter-chat	POST	3000	Grounded AI Recruiter query layer explaining candidate rankings and skill comparisons.

6. LOCALHOST DEPLOYMENT & TESTING

Backend (FastAPI): cd backend && ./run_backend.sh → Runs on **http://127.0.0.1:8000** (Interactive Swagger at /docs)
Frontend (React + Vite): cd frontend && ./run_frontend.sh → Runs on **http://localhost:3000** (Vite + Express Proxy)